An attained eventual reality threshold for a growing quartic arithmetic transform

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Abstract

For the growing quartic cosine transform of the arithmetic theta kernel, we prove that the set of eventual split-real polynomial starting times is a closed upper ray with finite endpoint strictly greater than eight. The proof includes uniform actual saddle estimates in both the moving transition region and its far complement, a finite exact Laguerre certificate, and a constructive polynomial approximation argument. A single radius confines all nonreal zeros at times at least one. Nonreal zeros approaching the endpoint have a subsequence colliding at a positive real zero of finite multiplicity at least two. All zeros are eventually real and simple. The meaning of eventual start is kept separate from pointwise-in-time reality, and no quadratic threshold or Riemann-hypothesis implication is asserted. The final statements are mapped to a frozen Lean development. A separate general divisor-product argument retains complex zeros and supplies a common Gaussian integration budget under an explicit zero-strip hypothesis; it gives no quartic propagation theorem. Their novelty remains unresolved.

1 The actual transform and its eventual starting times

For $u \geq 0$ define the arithmetic kernel

$$K(u) = \sum_{n=1}^{\infty} (2\pi^2 n^4 e^{9u} - 3\pi n^2 e^{5u}) e^{-\pi n^2 e^{4u}}, \tag{1}$$

and, for real t and complex z, put

$$H_t(z) = \int_0^{\infty} e^{tu^4} K(u) \cos(zu)\, du. \tag{2}$$

The sign of the quartic tilt is positive. It is neither the decaying quartic heat kernel nor the classical quadratic deformation. Let $\mathsf{LP}(t)$ mean that real polynomials splitting into real linear factors converge to H_t locally uniformly on the entire complex plane. Define

$$\mathcal{S} = \{T \in \mathbb{R} : \mathsf{LP}(t) \text{ for every } t \geq T\}, \qquad A = \inf \mathcal{S}. \tag{3}$$

The infimum becomes an ordinary finite real number once nonemptiness and a lower bound are proved.

Theorem 1.1. *The set $\mathcal{S}$ is nonempty and equals $[A, \infty)$ for a finite $A > 8$. For every $t \geq A$, all complex zeros of H_t are real. For every $\varepsilon > 0$ there is a time $s \in (A - \varepsilon, A)$ at which H_s has a nonreal zero. There is one finite radius containing every nonreal zero at every time $t \geq 1$. Some sequence of nonreal zeros at times below A converges to a positive real zero of H_A of finite multiplicity at least two. For every sufficiently large time, all zeros are real and simple.*

The last statement is eventual simplicity, not simplicity at every $t > A$. The eventual-start set in (3) must also be distinguished from $\{t : \mathsf{LP}(t)\}$. Equality of the former with $[A, \infty)$ does not exclude earlier individual reality times. No exact double multiplicity or continuously tracked zero path is asserted.

The classical quadratic theory uses a real-zero propagation property of a quadratic multiplier. De Bruijn's and Newman's primary work supplies the relevant comparison, but it does not permit that property to be silently applied to e^{tu^4}. The whole-plane statement here requires actual quartic estimates, including the region beyond every fixed frequency window. It has no proved implication for the Riemann hypothesis.

2 Arithmetic decay, entireness and time continuity

All operations in (2) concern an actual Lebesgue integral. We first verify its domains.

Lemma 2.1. *The kernel is continuous and strictly positive on $[0, \infty)$. There is $C > 0$ such that*

$$K(u) \le C \exp\bigl(9u - \tfrac{\pi}{2} e^{4u}\bigr), \qquad u \ge 0. \tag{4}$$

For every integer $k \ge 0$ and real T, B, $u^k e^{Tu^4 + Bu} K(u)$ is integrable on that half-line.

Proof. Each summand in (1) factors as

$$\pi n^2 e^{5u} (2\pi n^2 e^{4u} - 3) e^{-\pi n^2 e^{4u}} > 0.$$

For $v \ge 1$ and $n \ge 1$, $n^2 v \ge (n^2 + v)/2$. Hence

$$n^m e^{-\pi n^2 v} \le n^m e^{-\pi n^2/2} e^{-\pi v/2}.$$

The first factors have a finite sum for $m = 2, 4$. Uniform convergence on compact u-intervals proves continuity, and the two bounds, with $e^{5u} \le e^{9u}$, prove (4). The positive first summand proves strict positivity.

Eventually $u^k \le e^u$ and $Tu^4 + (B + 11)u \le (\pi/2) e^{4u}$. Substitution into (4) bounds the tilted moment by Ce^{-u}. Continuity handles its initial compact interval. □

Proposition 2.2. *The function H_t is entire, even and conjugation compatible, and $H_t(0) > 0$. Its spatial derivatives are jointly continuous in (t, z). Time convergence is uniform on every compact complex set. For each fixed real t there is $C_t > 0$ with*

$$|H_t(z)| \le C_t e^{|z|^{3/2}}. \tag{5}$$

Proof. On a compact time–space set, the kth derivative of the exponential integrand is dominated by $u^k e^{Tu^4 + Bu} K(u)$ for fixed finite T, B. Lemma 2.1 permits differentiation and continuity under the integral. The two exponentials defining cosine give the entire derivatives. Its elementary symmetries give evenness and conjugation, while positivity gives $H_t(0) > 0$. Joint continuity on a compact time interval times a compact complex set is uniform continuity, proving the compact-time claim.

For $R, u \ge 0$, $Ru \le u^3 + R\sqrt{R}$: split at $u = \sqrt{R}$. Also $u^3 \le u^4 + 1$. Using $|\cos(zu)| \le e^{|z|u}$ with $R = |z|$ gives

$$|H_t(z)| \le e^{1 + R^{3/2}} \int_0^\infty e^{(t+1)u^4} K(u)\, du.$$

The last integral is finite and positive. This proves the specific subquadratic bound (5). □

3 Real zeros and actual split-polynomial limits

For this even growth class, all-real zeros give actual polynomial approximation with its normalization restored. We give a finite-disk proof rather than assume a converse to a theorem about limits of zeros.

Lemma 3.1. *Let f be entire and even, with $f(0) \in \mathbb{R} \setminus \{0\}$, and suppose $|f(z)| \le Ce^{|z|^{3/2}}$. If every zero of f is real, then f is a locally uniform limit on $\mathbb{C}$ of real polynomials splitting over $\mathbb{R}$.*

Proof. Choose radii $R \to \infty$ with no zero on their boundary; isolated zeros exclude only countably many radii. List the finitely many zeros a in $|z| < R$, with multiplicity, and set

$$B_R(z) = \prod_{|a|<R} \frac{R(z-a)}{R^2 - \overline{a}z}.$$

No a is zero. The denominators are nonzero on the closed disk, and each factor has modulus one on its boundary. The quotient $h_R = f/B_R$ extends across all zeros to a zero-free holomorphic function on the disk, continuous and holomorphic near its boundary. The maximum modulus principle gives

$$|h_R(z)| \le M_f(R) \le Ce^{R^{3/2}}, \qquad |h_R(0)| \ge |f(0)|,$$

the last inequality using $|B_R(0)| = \prod |a|/R \le 1$.

The zero multiset occurs in pairs $a, -a$ with equal multiplicities. Each paired factor is even, so B_R and h_R are even. Choose the analytic logarithm g_R on the disk satisfying $e^{g_R} = h_R/h_R(0)$ and $g_R(0) = 0$. Uniqueness of that normalized logarithm makes it even. For some positive $A_R = O(1 + R^{3/2})$ we have $\operatorname{Re} g_R \le A_R$.

Here is the needed logarithm estimate explicitly. The function

$$v_R = \frac{g_R}{2A_R - g_R}$$

is holomorphic and bounded in modulus by one, because

$$|2A_R - g_R|^2 - |g_R|^2 = 4A_R(A_R - \operatorname{Re} g_R) \ge 0.$$

It is even and vanishes at zero. The Schwarz lemma applied with this double zero gives $|v_R(z)| \le (|z|/R)^2$. Solving for g_R yields

$$|g_R(z)| \le \frac{2A_R(|z|/R)^2}{1 - (|z|/R)^2}.$$

On each fixed compact disk this tends to zero as $R \to \infty$. Consequently $h_R/h_R(0) \to 1$ locally uniformly.

We also need a zero count to control the denominator product. Choose a zero-free boundary radius $\rho \in (2R, 3R)$. Harmonicity of $\log|h_\rho|$ and its boundary values give

$$\sum_{|a|<\rho} \log \frac{\rho}{|a|} \le \log M_f(\rho) - \log|f(0)|.$$

Indeed this follows by taking the boundary mean of $\log|h_\rho| = \log|f|$ and using $f(0) = B_\rho(0)h_\rho(0)$. Every zero in $|a| < R$ contributes at least $\log 2$, so its counting function satisfies $n_f(R) = O(1 + R^{3/2})$.

Now normalize the finite product:

$$\frac{B_R(z)}{B_R(0)} = \frac{N_R(z)}{D_R(z)}, \qquad N_R(z) = \prod_{|a|<R} (1 - z/a), \quad D_R(z) = \prod_{|a|<R} (1 - \overline{a}z/R^2).$$

Pairing $a, -a$ turns D_R into a product of $1 - \overline{a}^2 z^2/R^4$. On $|z| \le r$, the sum of the absolute pair corrections is at most $n_f(R)r^2/(2R^2) = O_r(R^{-1/2})$. The finite-product inequality $|\prod_j (1 - w_j) - 1| \le \exp(\sum_j |w_j|) - 1$ therefore gives $D_R \to 1$ locally uniformly. The exact identity

$$\frac{f}{f(0)} = \frac{B_R}{B_R(0)} \frac{h_R}{h_R(0)}$$

implies $N_R \to f/f(0)$ locally uniformly. Since all a are real, $f(0)N_R$ are the required split-real polynomials. □

Lemma 3.2. *A nonzero entire locally uniform limit of split-real polynomials has only real zeros.*

Proof. For every real root r, $|x + iy - r|$ is nondecreasing in $|y|$. Multiplying its linear factors gives $|P(x + ia)| \le |P(x + ib)|$ whenever $a^2 \le b^2$. The inequality survives the pointwise limits. A nonreal zero $x + iy$ of the limit would force zeros at all $x + iy/(n + 1)$, accumulating at x. The identity theorem would make the limit zero everywhere, a contradiction. □

Corollary 3.3. *For every real t, $\mathsf{LP}(t)$ holds if and only if every zero of the actual H_t is real.*

Proof. Use Proposition 2.2 and the preceding two lemmas. □

4 An actual arithmetic obstruction at time eight

This section records a computer-assisted finite certificate, with analytic errors kept separate from the finite checks. Let $x_0 = 632083/10000$ and $a = 8/5$. Let K_4 be the first four terms of (1), and put

$$c_0(u) = \cos(x_0 u), \quad c_1(u) = -u \sin(x_0 u), \quad c_2(u) = -u^2 \cos(x_0 u), \quad F_k(u) = e^{8u^4} K_4(u) c_k(u).$$

Lemma 4.1. *For $k = 0, 1, 2$,*

$$\left| H_8^{(k)}(x_0) - \int_0^a F_k(u)\, du \right| \le 10^{-16}.$$

Proof. On $[0, a]$ we have $u^k \le 3$. The positive exponential series gives $e^{4u} \ge 1 + 4u + (4u)^4/24$, hence $3n^2(e^{4u} - 1) \ge 9u + 8u^4$ for $n \ge 1$. Using $3 < \pi < 4$, the omitted positive kernel terms satisfy

$$\begin{aligned} u^k e^{8u^4} (K - K_4)(u) &\le 96 \sum_{n \ge 5} n^4 e^{-3n^2} \le 192 \sum_{n \ge 5} e^{-2n^2} \\ &< 384 e^{-50} < 384 (2/5)^{50} < 10^{-17}. \end{aligned}$$

Here $n^4 \le 2e^{n^2}$ follows from the exponential series, and $2(5 + j)^2 \ge 50 + 20j$ gives the geometric sum bound. Also $e > 5/2$.

The constant in (4) may be taken at most 176: for $m = 2, 4$,

$$\sum_{n\geq 1} n^m e^{-\pi n^2/2} \leq 2\sum_{n\geq 1} e^{-n^2/2} \leq \frac{2}{e^{1/2}-1} \leq 4,$$

and $2\pi^2 \cdot 4 + 3\pi \cdot 4 \leq 176$. For $u \geq a$, the functions $u^4 e^{-4u}$ and ue^{-4u} decrease. The rational inequality

$$8a^4 + 111a < \tfrac{3}{2}(5/2)^6 < \tfrac{3}{2}e^{4a}$$

therefore gives $8u^4 + 111u \leq (3/2)e^{4u}$ for all $u \geq a$. Since $u^k \leq e^{2u}$, the full tail integrand is bounded by $176e^{-100u}$. Its integral is at most $(176/100)e^{-160} < (176/100)2^{-160}$. Adding the two errors gives $a10^{-17} + (176/100)2^{-160} < 10^{-16}$. □

Divide $[0, a]$ into 160 cells, with centers $c_j = (2j+1)/200$ and half-width $h = 1/200$. Let $T_{j,k}$ be the degree-15 Taylor polynomial of F_k about c_j and define

$$P_k = \sum_{j=0}^{159} \int_{c_j-h}^{c_j+h} T_{j,k}(u)\, du.$$

Only its eight even coefficients contribute. The normalized sixteenth derivative bound on each cell is $|F_k^{(16)}(u)/16!| \leq 10^{22}$. Consequently

$$\left|\int_0^a F_k - P_k\right| \leq 160\frac{2\cdot 10^{22} h^{17}}{17} \leq \frac{16}{10^{15}}, \qquad |H_8^{(k)}(x_0) - P_k| \leq \frac{161}{10^{16}}.$$

The frozen exact certificate checks the derivative bound on all 160 whole cells and all 480 centered integral enclosures. At common denominator 10^{22}, its summed endpoint numerators are

k	lower	upper
0	−1641482419995	−1641482419749
1	3656317813	3656318043
2	−1460002952576	−1460002952347.

The full cell table and an independent exact rational re-sum are saved with the source dossier. The interval calculations use outward dyadic rounding at scale 2^{192}, 48-term range-reduced exponential and trigonometric series, and the normalized coefficient recursion

$$B_0 = e^{\phi}, \qquad B_n = \frac{1}{n}\sum_{j=1}^{n} jA_jB_{n-j}, \quad A_j = \phi^{(j)}/j!, \quad B_j = (e^{\phi})^{(j)}/j!.$$

The phases $8u^4 + bu - \pi n^2 e^{4u}$ retain $b = 9, 5$ and $n = 1, \ldots, 4$ separately. Every finite inclusion is proved by kernel reduction in the audited Lean source; this is an explicit computer-assisted step, not a floating-point probe.

Proposition 4.2. *At x_0 the actual transform satisfies*

$$H_8'(x_0)^2 - H_8(x_0)H_8''(x_0) < -2\cdot 10^{-20}. \tag{6}$$

There is an open neighborhood of 8 in which every time has a nonreal actual zero. Every eventual start is bounded below by some fixed number strictly greater than 8.

Proof. Enlarge the three table intervals by $161/10^{16}$. Their channel-zero and channel-two upper bounds are negative. Let their negatives be $b_0, b_2 > 0$ and let b_1 be the enlarged absolute channel-one bound. Exact rational arithmetic gives $b_1^2 - b_0 b_2 < -2 \cdot 10^{-20}$, proving (6).

Every real split polynomial satisfies $P'^2 - PP'' \geq 0$ on the real axis. Away from its roots this follows by dividing by P^2 and summing the positive terms $m_r/(x-r)^2$; continuity covers its roots. Locally uniform entire convergence also gives convergence of the first two derivatives by the Cauchy formula. Thus (6) excludes $\mathsf{LP}(8)$. Joint derivative continuity preserves the strict negative sign in an open time interval about 8, and Corollary 3.3 supplies its nonreal zeros. A fixed bad time in that interval strictly above 8 excludes every eventual start below or equal to it. □

5 The theta strip and its actual logarithmic derivative

The large-frequency analysis takes place in $\mathcal{U} = \{w : |\operatorname{Im} w| < \pi/8\}$. The same series as in (1) defines a holomorphic kernel there. All constants denoted $C, c > 0$ in the next two sections are absolute unless an explicit subscript records a fixed dependence. Their values may change between inequalities. A uniform assertion as $a \to \infty$ means that one cutoff in a works for every other parameter satisfying the displayed hypotheses.

Lemma 5.1. *The complex kernel is even and conjugation compatible on $\mathcal{U}$. If $w = a + ib \in \mathcal{U}$, $a \geq 0$, $L = e^{4a}$ and $V = \operatorname{Re} e^{4w}$, then*

$$K(w) = 2\pi^2 e^{9w - \pi e^{4w}}(1 + \rho(w)), \qquad |\rho^{(j)}(w)| \leq C_j\big(e^{-4a} + (1+L)^j e^{-3\pi V}\big) \tag{7}$$

for $0 \leq j \leq 4$ when $V \geq 1$. For $s \geq 0$ and $c_b = \cos(4b) > 0$,

$$|K(s + ib)| \leq C c_b^{-3} e^{-s}. \tag{8}$$

Proof. Normal convergence of the theta series and all its derivatives holds on compact subsets where $\operatorname{Re} e^{4w} > 0$. Conjugation follows termwise. To check evenness, write $\vartheta(v) = \sum_{n \in \mathbb{Z}} e^{-\pi n^2 v}$ and $F(w) = e^w(\vartheta(e^{4w}) - 1)/2$. The Gaussian theta identity $\vartheta(v) = v^{-1/2}\vartheta(1/v)$, with the analytic right-half-plane square root, gives $F(-w) = F(w) + \sinh w$ on $\mathcal{U}$. This identity is the Gaussian Poisson summation formula; equivalently it is the zero-argument modular theta identity in [3]. Direct differentiation gives $(d^2/dw^2 - 1)F = 8K$. Applying this operator to the reflection identity proves $K(-w) = K(w)$.

Factor out the $n = 1$ leading exponential. The remaining factor is

$$1 - \frac{3}{2\pi} e^{-4w} + \sum_{n \geq 2} \left(n^4 - \frac{3}{2\pi} n^2 e^{-4w}\right) e^{-\pi(n^2-1)e^{4w}}.$$

Each of its first four derivatives is a finite linear combination of terms bounded by a constant times e^{-4a} or $(1+L)^j n^{4+2j} e^{-\pi(n^2-1)V}$. For $V \geq 1$, dividing the last sum by $e^{-3\pi V}$ leaves a sum bounded by $\sum_{n \geq 2} n^{4+2j} e^{-\pi(n^2-4)} < \infty$. This proves (7), including its derivative bounds.

For every $V > 0$ and $m = 2, 4$, $\sum_{n \geq 1} n^m e^{-\pi n^2 V} \leq C_m V^{-(m+1)/2}$. Indeed $n^m e^{-\pi n^2 V/2} \leq C_m V^{-m/2}$, and the remaining decreasing Gaussian sum is bounded by its integral from zero. Substitute $V = e^{4s} c_b$ in the two kernel terms. They are bounded by $Ce^{-s}(c_b^{-5/2} + c_b^{-3/2})$, which implies (8) since $0 < c_b \leq 1$. □

Write

$$\sigma_t(w) = 4tw^3 + \frac{K'(w)}{K(w)}, \qquad W_{t,z}(w) = e^{tw^4+izw}K(w)$$

where the denominator is nonzero. Under $V \geq c_0 ta^3$, $t \geq 1$ and large a, formula (7) gives

$$\sigma_t(w) = 4tw^3 + 9 - 4\pi e^{4w} + \epsilon(w), \qquad |\epsilon^{(j)}(w)| \leq Ce^{-4a} \quad (0 \leq j \leq 3), \tag{9}$$

for any fixed $c_0 > 0$. To see the uniformity, the other error terms are at most $C\exp(16a - ca^3)$, which is eventually bounded by e^{-4a}. The same bound holds along horizontal shifts $w + s$, $s \geq -1$, after replacing the right side by Ce^{-4a}: the potentially growing terms have the form $\exp(16a + 16s - cta^3e^{4s})$, whose maximum for $s \geq -1$ occurs at the left endpoint once a is large. In particular the kernel is nonzero on those horizontal segments.

The distinction between the full slope σ_t and its leading model $4tw^3 + 9 - 4\pi e^{4w}$ remains essential. Stationarity below always refers to the full slope.

6 A uniform horizontal saddle estimate

Fix $c_0 = 1/1000$. Call (t, z, v) admissible when

$$\begin{gathered} t \geq 1, \quad v = a + ib \in \mathcal{U}, \quad a \text{ sufficiently large}, \quad y = \operatorname{Im} z \leq 1, \\ \sigma_t(v) + iz = 0, \qquad \operatorname{Re} e^{4v} \geq c_0 ta^3. \end{gathered} \tag{10}$$

The last condition is verified in the saddle constructions; it prevents circular use of a theta asymptotic at an unknown point. Set $Y = \max(0, -y)$, $Q = ta^3 + Y$ and $L = e^{4a}$.

Lemma 6.1. *Uniformly for admissible saddles,*

$$cQ \leq \operatorname{Re} e^{4v} \leq CQ, \quad cL \leq |\beta| \leq CL, \quad cQ \leq \operatorname{Re}\beta \leq CQ, \qquad \beta = -\sigma_t'(v)/2. \tag{11}$$

For real $s \geq -1$,

$$\operatorname{Re}\sigma_t'(v + s) \leq -cQ. \tag{12}$$

On each sufficiently small fixed horizontal interval about zero, $|\sigma_t''(v + s)| \leq CL$.

Proof. Take the real part of full stationarity in (9):

$$4\pi \operatorname{Re} e^{4v} = 4t(a^3 - 3ab^2) + 9 - y + \operatorname{Re}\epsilon(v). \tag{13}$$

Since $|b| < \pi/8 < 1/2$, a is large and $y \leq 1$, this is comparable above and below to $ta^3 + Y$. It also gives $ta^3 \leq CL$. In

$$\sigma_t'(v) = 12tv^2 - 16\pi e^{4v} + \epsilon'(v),$$

the first term has norm at most CL/a, whereas the exponential term has norm $16\pi L$. This proves the norm bounds for β. Taking real parts and using (13) gives its real-part bounds, since $ta^2 = o(ta^3)$ uniformly.

For $s \geq -1$, $(a + s)^2 \leq Ca^2e^{2s}$ and $\operatorname{Re} e^{4(v+s)} = e^{4s}\operatorname{Re} e^{4v}$. The positive polynomial part of $\operatorname{Re}\sigma_t'(v + s)$ is therefore at most $(C/a)Qe^{2s}$. The negative exponential part is at most $-cQe^{4s}$. Since $e^{2s} \leq e^2e^{4s}$ on this interval, the negative term dominates for large a. The uniform error (9) is negligible as well, proving (12). Finally $\sigma_t''(w) = 24tw - 64\pi e^{4w} + \epsilon''(w)$; the asserted small-window bound follows from $ta^3 \leq CL$. □

For fixed admissible parameters, the shifted horizontal integral is absolutely convergent: its strip margin is positive and the double exponential dominates the quartic term at infinity. Put

$$I_{t,z}(b) = \int_0^\infty W_{t,z}(s+ib)\,ds, \qquad G(\beta) = \int_{\mathbb{R}} e^{-\beta s^2}\,ds.$$

Proposition 6.2. *There is a function $\eta(a) \to 0$ such that, uniformly for all admissible saddles,*

$$\left| \frac{I_{t,z}(b)}{W_{t,z}(v)G(\beta)} - 1 \right| \le \eta(a). \tag{14}$$

Also

$$\int_0^\infty |W_{t,z}(s+ib)|\,ds \le C|W_{t,z}(v)|/\sqrt{Q}. \tag{15}$$

Proof. On the horizontal interval $s \ge -1$, define

$$P(s) = \int_0^s (\sigma_t(v+r) + iz)\,dr.$$

The elementary scalar ODE for the weight gives $W_{t,z}(v+s) = W_{t,z}(v)e^{P(s)}$. This follows by differentiating $W_{t,z}(v+s)e^{-P(s)}$, whose derivative is zero; no complex neighborhood as wide as the horizontal interval is assumed. Full stationarity gives $P'(0) = 0$. Lemma 6.1 yields

$$\operatorname{Re} P(s) \le -cQs^2 \quad (s \ge -1), \qquad P(s) = -\beta s^2 + O(L|s|^3) \qquad (|s| \le \rho), \tag{16}$$

where $\rho > 0$ is a sufficiently small absolute constant. The second estimate is real-parameter Taylor's theorem with the uniform horizontal third-derivative bound.

For $\operatorname{Re}\beta > 0$, the complex Gaussian formula gives $G(\beta) = \sqrt{\pi}/\sqrt{\beta}$, with its right-half-plane branch. One direct verification differentiates under the absolutely dominated integral and integrates the derivative of $se^{-\beta s^2}$: it obtains $G' = -G/(2\beta)$. Thus $\sqrt{\beta}\,G$ is constant on the right half-plane, and its value on the positive real axis is $\sqrt{\pi}$. Hence $|G(\beta)| \asymp L^{-1/2}$ by (11). Its two tails beyond $r > 0$ satisfy

$$\left| \int_r^\infty e^{-\beta s^2}\,ds \right| + \left| \int_{-\infty}^{-r} e^{-\beta s^2}\,ds \right| \le \frac{2}{|\beta| r}. \tag{17}$$

Integration by parts gives, for the positive tail, $e^{-\beta r^2}/(2\beta r) - (2\beta)^{-1}\int_r^\infty e^{-\beta s^2} s^{-2}\,ds$; taking absolute values proves the bound, and the negative tail is identical. This estimate uses $|\beta|$, not a false comparison of its real part with its modulus.

Take $r = L^{-7/16}$. On $[-r, r]$ the remainder in (16) is eventually at most one. The inequality $|e^w - 1| \le e^{|w|}|w|$ bounds the integral difference from $e^{-\beta s^2}$ by CLr^4. Relative to G, this is $O(L^{3/2}r^4) = O(L^{-1/4})$. The Gaussian tails (17), after normalization, are $O(L^{-1/2}r^{-1}) = O(L^{-1/16})$.

It remains to estimate the actual annuli. For $r \le |s| \le \rho$, reduce ρ if necessary so that

$$|P'(s)| \ge cL|s|, \qquad |P''(s)| \le CL.$$

Indeed $P'(s) = -2\beta s + O(Ls^2)$ and $|\beta| \ge cL$. On either annulus integrate $(e^P)'/P'$ by parts. The boundary terms are bounded by $C/(Lr)$. The integral of $|e^P P''/(P')^2|$ is at most

$$C\int_r^\rho \frac{ds}{Ls^2} \le \frac{C}{Lr}.$$

We used $|e^P| \le 1$ from (16). The relative annular error is again $O(L^{-1/16})$.

The remaining interval $[-1,-\rho]$ and right tail $[\rho,\infty)$ are bounded absolutely by $Ce^{-cQ\rho^2}$ and $Ce^{-cQ\rho^2}/(Q\rho)$ respectively. After multiplication by $\sqrt{L} = e^{2a}$, both tend to zero uniformly since $Q \ge a^3$.

Finally consider the physical prefix $0 \le s \le a-1$. Write $z = X + iy$. Equation (13) gives $c_b = \operatorname{Re} e^{4v}/L \ge cQ/L$. Thus (8) and $-ys \le Ys$ give

$$|W_{t,z}(s+ib)| \le Ce^{12a} Q^{-3} e^{t(a-1)^4 + Ct + Y(a-1) - Xb}.$$

The real quartic term on the line also contains $-6ts^2b^2 + tb^4 \le Ct$, which can be included in the exponent. At the center, (7) and (13) give

$$|W_{t,z}(v)| \ge ce^{ta^4 - ta^3 - Cta^2 + 9a + Y(a-1/4) - Xb - Ca}.$$

For $0 < y \le 1$ the replacement of $-ya$ by this lower bound costs only Ca. Subtract the two exponents. The polynomial $t((a-1)^4 - a^4 + a^3 + Ca^2 + C)$ is at most $-cta^3$ for large a, and the Y term is $-3Y/4$. Therefore

$$\frac{\int_0^{a-1} |W_{t,z}(s+ib)|\, ds}{|W_{t,z}(v)|} \le Ca\, e^{-cQ+Ca}. \tag{18}$$

It too vanishes after Gaussian normalization. The central, annular and three complementary bounds prove (14). One may take a constant multiple of $L^{-1/4} + L^{-1/16} + (a+1)e^{-ca^3+Ca}$ for η, after enlarging the constants. Absolute integration of e^{-cQs^2} on $[-1,\infty)$, together with (18), proves (15). □

This proof retains the actual horizontal slope and curvature. The formal source often compares them to their leading model, carrying its small linear mismatch explicitly. The present grouping uses the same arithmetic estimates at the actual saddle and proves the resulting bound directly. It does not attribute this regrouping to an additional named Lean theorem.

7 Signed contours, analytic pairing and exact local zeros

The horizontal estimate concerns one half-line contribution. We now relate it to the actual cosine integral, retaining the endpoint terms when the two heights differ.

Lemma 7.1. *For real h, k with $|h|, |k| < \pi/8$,*

$$H_t(z) = \tfrac{1}{2}\Big(I_{t,z}(h) + \overline{I_{t,\bar z}(k)} + i\int_k^h W_{t,z}(iy)\, dy \Big). \tag{19}$$

In particular, choosing mirrored heights for the original $z, -z$ contributions gives the exact identity

$$H_t(z) = \tfrac{1}{2}\big(I_{t,z}(h) + I_{t,-z}(-h)\big). \tag{20}$$

Proof. Integrate $W_{t,z}$ around the rectangle with vertices $0, R, R+ih, ih$. It is holomorphic on a neighborhood of this rectangle in the theta strip. Its right vertical edge tends to zero as $R \to \infty$: throughout that edge $\operatorname{Re} e^{4(R+iy)} \ge e^{4R} \min_{|v|\le|h|} \cos(4v) > 0$, and the theta decay dominates

every quartic and linear factor. The two horizontal integrals are absolutely convergent for the same reason. Their orientations therefore give

$$J_t(z) := \int_0^\infty W_{t,z}(u)\,du = I_{t,z}(h) + i\int_0^h W_{t,z}(iy)\,dy.$$

Apply this also to $J_t(-z)$ at height $-k$. Kernel evenness gives $W_{t,-z}(-iy) = \overline{W_{t,z}(iy)}$, so its vertical term is the negative integral from 0 to k. Conjugation gives $I_{t,-z}(-k) = \overline{I_{t,\bar{z}}(k)}$. Adding and dividing by two proves (19). For $k = h$ the vertical integral is zero, giving (20) in the original opposite-frequency notation. □

Lemma 7.2. *For an admissible saddle with $y = \operatorname{Im} z \le 0$,*

$$\frac{|I_{t,-z}(-b)|}{|W_{t,z}(v)G(\beta)|} \le C\sqrt{L/Q}\,e^{2y(a-1)} + o(1), \tag{21}$$

uniformly as $a \to \infty$. Consequently no zero can occur at such an admissible frequency with $y \le -1$, for large a.

Proof. At the same physical horizontal coordinate $s \ge 0$, conjugation of the kernel gives the exact modulus ratio

$$|W_{t,-z}(s - ib)| = |W_{t,z}(s + ib)|e^{2ys}.$$

On $s \ge a - 1$, use $e^{2ys} \le e^{2y(a-1)}$ and (15). On $0 \le s \le a - 1$, use $e^{2ys} \le 1$ and (18). Division by $|G| \asymp L^{-1/2}$ proves (21). If $y \le -1$, then $\sqrt{L}\,e^{2y(a-1)} \le e^{2a}e^{-2(a-1)} = e^2$. Since $Q \ge a^3$, the right side tends to zero. The direct contribution in (20), normalized by the same nonzero amplitude, tends to one by Proposition 6.2. Their sum cannot vanish. □

The remaining strip needs exact zero control. A real conjugation-compatible function close to cosine can be shown injective on the small disk containing a proposed conjugate zero pair. The buffer in the next lemma is what permits the derivative estimate.

Lemma 7.3. *Let $a \in \mathbb{R}$ and let f be holomorphic near the closed disk $|q-a| \le 1$, with $f(\overline{q}) = \overline{f(q)}$ there. If $|f(q) - \cos q| \le \delta = 1/1000$ on that disk, every zero in $|q - a| \le \delta$ is real and simple.*

Proof. At a proposed zero q_0, $|\cos q_0| \le \delta$. Since $|\sin q| \le e^{|\operatorname{Im} q|} < 3$ on the unit disk, integration along the segment gives $|\cos a| \le \delta + 3\delta = 1/250$. Thus $|\sin a| \ge 1/2$. Set $d = -\sin a$. For $|q-a| \le \delta$, the circle of radius $1/2$ about q stays inside the unit disk. Cauchy's derivative estimate applied to $f - \cos$ gives $|f'(q) + \sin q| \le 2\delta$. Also $|\sin q - \sin a| \le 3\delta$, so $|f'(q) - d| \le 5\delta \le |d|/2$ on the small disk.

Integrate $f' - d$ on the segment between any two points of that convex disk. The reverse triangle inequality gives

$$|f(q) - f(p)| \ge \tfrac{1}{2}|d|\,|q - p|.$$

The conjugate of q_0 is another zero in the disk, so the inequality forces $q_0 = \overline{q}_0$. The derivative bound also gives $|f'(q_0)| \ge |d|/2 > 0$. □

Lemma 7.4. *Let g, E, Φ be holomorphic on $|z - x| < r$, where $x \in \mathbb{R}$, with E nonzero and all three functions conjugation compatible. Suppose $s > 0$, $D \ge 0$,*

$$D/s \le 10^{-6}, \quad 1/s \le r/2, \quad |\Phi'(z) - s| \le D,$$

and

$$|g(z)/E(z) - \cos\Phi(z)| \le 10^{-6}\cosh(\operatorname{Im}\Phi(z)). \tag{22}$$

Every zero z with $|z - x| < r/2$ is real and simple.

Proof. Put $p = \operatorname{Re} z$, $a = \Phi(p) \in \mathbb{R}$, $\ell(q) = p + (q-a)/s$ and $f(q) = g(\ell(q))/E(\ell(q))$. The closed unit disk about a maps strictly inside the original disk, by the buffer and $|p-x| < r/2$. Integration of $\Phi' - s$ gives $|\Phi(\ell(q)) - q| \le D/s \le \varepsilon = 10^{-6}$. Both arguments are in the disk of radius two about a; there $|\sin| < 9$ and $\cosh(\operatorname{Im}\Phi) \le 9$. Thus (22) implies $|f(q) - \cos q| \le 18\varepsilon < 1/1000$.

At the zero, writing $\Phi(z) = u + iv$ gives $|\cos(u+iv)|^2 = \cos^2 u + \sinh^2 v$. Equation (22) therefore implies $|\tanh v| \le \varepsilon$, hence $|v| \le \operatorname{arctanh}\varepsilon \le 2\varepsilon$. On the vertical segment from p to z, $\operatorname{Re}\Phi' \ge s - D \ge s/2$. Since $\Phi(p)$ is real, integration gives $|\operatorname{Im}\Phi(z)| \ge s|\operatorname{Im} z|/2$. Consequently $q_0 = a + s(z-p)$ is a zero of f with $|q_0 - a| = s|\operatorname{Im} z| \le 4\varepsilon < 1/1000$. Lemma 7.3 makes q_0 real and simple. Thus z is real, and $f'(q_0) = g'(z)/(sE(z)) \ne 0$ proves its simplicity. □

Proposition 7.5. *Suppose a real frequency x has an admissible actual saddle $v_0 = a + ib$, with large a. Then every zero $z = x + iy$ of H_t with $|y| < 1$ is real and simple.*

Proof. We first construct the local analytic quantities rather than differentiate a varying principal argument. Lemma 6.1 gives $|\sigma_t'(v_0)| \asymp L$. The distance of v_0 from the theta-strip boundary is at least cQ/L: indeed $\cos(4b) \ge cQ/L$ and the distance is comparable to this cosine near either boundary. Since $Q \ge a^3$, a disk of radius C/L lies inside the strip for any fixed C and large a. On this disk $|\sigma_t''| \le CL$, by the differentiated series estimate (7); the real-part margin changes by only a bounded amount.

For $|z - x| < 2$, solve $\sigma_t(v) + iz = 0$ by the contraction

$$v \longmapsto v - \frac{\sigma_t(v) + iz}{\sigma_t'(v_0)}$$

on a disk $|v - v_0| \le C/L$ with a sufficiently large fixed C. Its center displacement is $O(1/L)$, and its derivative is $O(1/L)$ there. It therefore maps the disk into itself and is contractive for large a. The fixed point $v(z)$ depends holomorphically on z, either by uniform iteration on smaller frequency disks or the analytic inverse theorem, and

$$v(z) = v_0 + O(1/L), \qquad v'(z) = -i/\sigma_t'(v(z)) = O(1/L). \tag{23}$$

For $|z - x| \le 1$ its imaginary frequency is at most one. The initial theta margin changes by only $O(1)$ and so first remains at least $(c_0/2)ta^3$, enough to use the series estimate (9). Taking real parts of its exact stationarity then gives (13) and restores the larger lower bound $\operatorname{Re} e^{4v(z)} \ge c_0 t(\operatorname{Re} v(z))^3$ for large a. Thus these saddles are admissible with the original fixed constant.

The actual amplitude

$$\mathcal{A}(z) = W_{t,z}(v(z))\sqrt{\pi/\beta(z)}, \qquad \beta(z) = -\sigma_t'(v(z))/2,$$

is nonzero and holomorphic. The square root is fixed by $\operatorname{Re}\beta > 0$. Stationarity cancels the derivative of the center in the exponential weight, giving

$$\frac{\mathcal{A}'}{\mathcal{A}} = iv(z) - \frac{\beta'(z)}{2\beta(z)} = iv(z) + O(1/L). \tag{24}$$

Here $\beta' = O(1)$ follows from $|\sigma_t''| \le CL$ and (23).

Choose an analytic P with $P(x) = 0$ and $\mathcal{A}(z) = \mathcal{A}(x)e^{P(z)}$. Put $P^*(z) = \overline{P(\overline{z})}$, $a_0 = \mathcal{A}(x)$ and

$$E(z) = |a_0|e^{(P(z)+P^*(z))/2}, \qquad \Phi(z) = \arg a_0 + \frac{P(z) - P^*(z)}{2i}.$$

These are holomorphic and conjugation compatible; E is nonzero. The argument of a_0 is a fixed real number. Direct expansion gives the exact identities

$$\begin{aligned} \tfrac{1}{2}(\mathcal{A}(z) + \overline{\mathcal{A}(\overline{z})}) &= E(z)\cos\Phi(z), \\ |\mathcal{A}(z)| + |\mathcal{A}(\overline{z})| &= 2|E(z)|\cosh(\operatorname{Im}\Phi(z)). \end{aligned} \tag{25}$$

Equations (23)–(24) also give $|\Phi'(z) - a| \le C/L$ on the unit frequency disk.

Apply (19) with heights $h = \operatorname{Im} v(z)$ and $k = \operatorname{Im} v(\overline{z})$. The two horizontal errors are at most $o(1)(|\mathcal{A}(z)| + |\mathcal{A}(\overline{z})|)$, uniformly. We must still bound the vertical residual. Its height gap is $O(|y|/L)$ by (23). Throughout the interval between the heights, the cosine margin is at least cQ/L: cosine is concave on $(-\pi/2, \pi/2)$, so its minimum on that interval is at an endpoint. Equation (8) at real coordinate zero bounds the kernel there by CL^3/Q^3. Also $|\operatorname{Re} z| \le CL$, as follows by taking norms in full stationarity at the center. Thus the vertical residual is at most

$$\exp(Ca + t - Xh + C), \qquad X = \operatorname{Re} z,$$

after absorbing its polynomial factors and using $|X||h - k| = O(1)$. On the other hand (13) at $v(z)$, with $|y| \le 1$, gives

$$|\mathcal{A}(z)| \ge \exp(ta^4 - Cta^3 - Ca - Xh).$$

Changing its real center by $O(1/L)$ changes only the stated constants. The ratio of the residual to this amplitude is at most $\exp(-cta^4 + Ca)$ for large a, hence tends to zero uniformly. It cannot be discarded as an exact cancellation, but it is negligible in this relative estimate.

By (25), we have now proved

$$|H_t(z)/E(z) - \cos\Phi(z)| \le o(1)\cosh(\operatorname{Im}\Phi(z)), \qquad |z - x| \le 1.$$

For large a replace $o(1)$ by 10^{-6}. If a proposed zero has $|y| < 1$, this inequality first gives $|\operatorname{Im}\Phi(z)| \le 2 \cdot 10^{-6}$ as in the preceding lemma. Vertical integration of Φ' gives $|\operatorname{Im}\Phi(z)| \ge (a - C/L)|y|$, so actually $|y| < 1/2$. Apply Lemma 7.4 on the unit disk, with $s = a$, $D = C/L$ and $r = 1$. Its buffer and scaled derivative conditions hold for large a. The zero is real and simple. □

8 Actual saddles in the transition and far regions

The preceding estimates do not assume that a saddle exists. We now supply that input in two overlapping frequency regimes. The distinction is quantitative: a fixed multiple of the moving real scale, and the region beyond that multiple.

Lemma 8.1. *For every sufficiently large t there is a positive real stationary center $u = u_t$ with $u_t \to \infty$ as $t \to \infty$. Writing $B = 4\pi e^{4u}$, it satisfies*

$$\sigma_t(u) = 0, \qquad B = 4tu^3 + 9 + O(e^{-4u}) \asymp tu^3. \tag{26}$$

Proof. The positive continuous function $e^{tu^4}K(u)$ tends to zero at infinity and has a global maximum on $[0, \infty)$. For each fixed $R \ge 0$, its maximum on $[0, R]$ is at most $e^{tR^4}\max_{[0,R]} K$ for $t \ge 0$. Its value at $R + 1$ is $e^{t(R+1)^4}K(R + 1)$, which eventually exceeds that bound. Thus every maximizing center escapes to infinity. At a positive center its ordinary derivative vanishes, so $4tu^3 + K'(u)/K(u) = 0$. The real-axis instance of (7) gives $K'(u)/K(u) = 9 - 4\pi e^{4u} + O(e^{-4u})$. This uses only the real kernel asymptotic, independent of t, and proves (26) without assuming it in advance. □

Proposition 8.2. *Fix $M \geq 0$. For all sufficiently large t, every zero in $|z| \leq MB$ is real and simple, with the same u, B as in Lemma 8.1.*

Proof. We construct saddles for all $|z| \leq MB$, $\operatorname{Im} z \leq 0$. The anchor

$$q_z = u + \tfrac{1}{4} \operatorname{Log}(1 + iz/B)$$

has $|q_z - u| \leq C_M$ and a fixed positive distance, depending only on M, from the boundary of $\mathcal{U}$. Indeed $\operatorname{Re}(1 + iz/B) \geq 1$ and $|1 + iz/B| \leq 1 + M$. On the disk $|w - q_z| \leq C_M/u$, all kernel errors and their derivatives in (9) are $O_M(e^{-4u})$. Equation (26) makes actual stationarity equivalent to the fixed-point equation

$$w = u + \tfrac{1}{4} \operatorname{Log}\left(1 + \frac{iz}{B} + \frac{4t(w^3 - u^3) + \epsilon(w) - \epsilon(u)}{B}\right). \tag{27}$$

The expression inside the logarithm remains in a right half-plane. Its perturbation from $1 + iz/B$ is $O_M(1/u)$, since $t/B = O(u^{-3})$. At the center the right side differs from q_z by $O_M(1/u)$, and its derivative in w is $O_M(tu^2/B) = O_M(1/u)$. Choose the fixed constant in the disk radius large enough for the center bound and then take u large. The map is a contraction of that disk into itself, giving an actual root $v(z)$ with

$$\operatorname{Re} v = u + O_M(1), \qquad \operatorname{Re} e^{4v} \asymp_M B, \qquad e^{4 \operatorname{Re} v} \asymp_M B.$$

In particular the theta margin is positive. Taking real parts of stationarity, as in (13), improves this to $\operatorname{Re} e^{4v} \geq c_0 t (\operatorname{Re} v)^3$ for large u. These saddles are admissible, and their ratio L/Q is bounded by a constant depending only on M.

Use Lemma 7.2 on any lower-half-plane frequency with $y \leq -A_M/u$. Since $\operatorname{Re} v - 1 \geq u/2$ for large u, its reflected relative amplitude is at most $C_M e^{-A_M} + o(1)$. Choose the fixed A_M so this is below $1/4$; the direct relative amplitude is within $1/4$ of one for large u. Equation (20) excludes such zeros. Conjugation excludes the corresponding upper region. Thus every zero in the transition disk has $|\operatorname{Im} z| < A_M/u < 1$.

At its real projection $x = \operatorname{Re} z$ we have $|x| \leq MB$; the same construction supplies a real-frequency admissible saddle. Proposition 7.5 now proves exact reality and simplicity of the zero. All cutoffs in this argument depend only on the fixed M, not on the sampled frequency. □

Lemma 8.3. *There is a sufficiently large absolute C_0 with the following property. Let $t \geq 1$, $\operatorname{Im} z \leq 0$, $X = |z|$ and*

$$\alpha = \tfrac{1}{4} \log(X/(4\pi)).$$

If α is sufficiently large and $X \geq C_0 t(\alpha + 1)^3$, there is an admissible actual saddle v with $\operatorname{Re} v = \alpha + O(1)$, uniformly in these parameters.

Proof. First solve the leading model, which is entire and has no theta-strip domain restriction. The anchor $q = \frac{1}{4} \operatorname{Log}(iz/(4\pi))$ has real part α and imaginary part in $[-\pi/8, \pi/8]$, since $\operatorname{Re}(iz) \geq 0$. Consider

$$w \longmapsto \tfrac{1}{4} \operatorname{Log}\left(\frac{iz + 4tw^3 + 9}{4\pi}\right)$$

on a disk about q of radius $Ct(\alpha + 1)^3/X$. Choose C_0 sufficiently large that this radius is below $1/100$. On this disk $\operatorname{Re}(w^3) \geq \alpha^3/2$ for large α, so the logarithm argument has positive real part. Its modulus is comparable to X, since the cubic perturbation has norm at most CX/C_0. The

center displacement is at most $Ct(\alpha+1)^3/X$ by $|\operatorname{Log}(1+\zeta)| \le 2|\zeta|$ for $|\zeta| \le 1/2$; its derivative is at most $Ct(\alpha+1)^2/X < 1/2$. Enlarging the fixed radius constant and then C_0 gives a contraction. Its fixed point w satisfies

$$4tw^3 + 9 - 4\pi e^{4w} = -iz, \quad |w-q| \le Ct(\alpha+1)^3/X.$$

Writing $w = a + ib$ and $Y = -\operatorname{Im} z \ge 0$, real parts give

$$4\pi \operatorname{Re} e^{4w} = 4t(a^3 - 3ab^2) + 9 + Y \ge cta^3 + Y.$$

We have $|b| < 1/2$, and positivity of this real part therefore places w strictly inside $|b| < \pi/8$. Also $e^{4a} \asymp X$.

On the disk $|v-w| \le 1/X$, the real theta margin changes by only $O(1)$, so remains at least $cta^3 + cY$. The error and its first derivative in (9) are $O(e^{-4a})$ there. The model derivative equals

$$12tw^2 - 16\pi e^{4w} = -4iz + t(12w^2 - 16w^3) - 36;$$

its norm is between cX and CX, because $ta^3 \le CX/C_0$. Its variation across this disk is $O(1)$. The Newton map using this fixed model derivative consequently has derivative $O(1/X)$ on the disk and center displacement $O(e^{-4a}/X)$. For large α it maps the disk into itself and is contractive. Its fixed point solves full stationarity. The retained positive real theta margin and $|\operatorname{Im} v| < 1/2$ keep it in the strip, and its margin exceeds $c_0 t(\operatorname{Re} v)^3$. It is the asserted admissible actual saddle. □

Proposition 8.4. *For a sufficiently large fixed constant C_1, if $t \ge 1$, $\alpha = \frac{1}{4}\log(|z|/(4\pi))$ is sufficiently large and*

$$|z| \ge C_1 t(\alpha+1)^3, \tag{28}$$

then every zero $H_t(z) = 0$ is real and simple. For each finite T, one radius works for every $1 \le t \le T$.

Proof. Choose C_1 at least twice the C_0 in the preceding lemma. If $\operatorname{Im} z \le -1$, its actual saddle and Lemma 7.2 exclude a zero. Conjugation excludes $\operatorname{Im} z \ge 1$. In the remaining strip, put $x = \operatorname{Re} z$. For large $|z|$ we have $|x| \ge |z|/2$, and $\alpha_x = \frac{1}{4}\log(|x|/(4\pi)) \le \alpha$. Thus $|x| \ge C_0 t(\alpha_x+1)^3$, with α_x large. At this real frequency the far-saddle construction applies. Proposition 7.5 proves that the proposed zero in $|\operatorname{Im} z| < 1$ is real and simple.

For $1 \le t \le T$, condition (28) and the lower bound on α hold for every sufficiently large $|z|$, uniformly in t, since $X/(1+\log X)^3 \to \infty$. If $T < 1$ the assertion is vacuous. □

9 Whole-plane assembly with one moving center

The bounded-ratio and far estimates cover all frequencies only after their scales are compared at the same real center.

Proposition 9.1. *For every sufficiently large t, every complex zero of H_t is real and simple.*

Proof. Use the actual maximizing center u and $B = 4\pi e^{4u}$ from Lemma 8.1. For $X = |z| \ge MB$, put $\alpha = \frac{1}{4}\log(X/(4\pi)) = u + d$. Then $d \ge \frac{1}{4}\log M$ and

$$\frac{t(\alpha+1)^3}{X} \le Ce^{-4d}\left(1 + \frac{d+1}{u}\right)^3 \le C(1+d)^3 e^{-4d}, \qquad u \ge 1.$$

Choose one fixed $M > 1$ so large that the last quantity is at most $1/C_1$ for every $d \geq \frac{1}{4}\log M$. This is possible by exponential domination. For late t, the far condition and lower bound on α then hold throughout $X \geq MB$. Proposition 8.4 treats that region. For this same fixed M and the same center u, Proposition 8.2 treats $X \leq MB$. The two regions cover $\mathbb{C}$. This proves the assertion without exchanging a fixed-time far limit with a late-time limit. □

Corollary 9.2. *There is $R < \infty$ such that every nonreal zero at every time $t \geq 1$ lies in $|z| < R$.*

Proof. Choose $T \geq 1$ after which Proposition 9.1 holds. On $[1, T]$ use the uniform radius from Proposition 8.4. Beyond T there are no nonreal zeros. Enlarge the radius to make it positive. □

10 Attainment and the multiple real zero

It remains to pass from eventual analytic control to the literal threshold in (3). No propagation property of the quartic flow is used in this passage.

Lemma 10.1. *If $s_n \to t$ and $\mathsf{LP}(s_n)$ for every n, then $\mathsf{LP}(t)$.*

Proof. For each n select one split-real polynomial Q_n within $1/(n+1)$ of H_{s_n} on $|z| \leq n$. On a fixed compact set, the error to H_t is eventually bounded by this quantity plus $\sup|H_{s_n} - H_t|$, which tends to zero by Proposition 2.2. Thus $Q_n \to H_t$ locally uniformly on the whole plane. □

Proof of Theorem 1.1. Proposition 9.1 and Corollary 3.3 make $\mathcal{S}$ nonempty. Proposition 4.2 bounds every start below by a fixed number greater than eight. Thus A is finite and $A > 8$. For each $t > A$, the definition of an infimum gives some $T \in \mathcal{S}$ with $T < t$, whence $\mathsf{LP}(t)$. Apply Lemma 10.1 to $t_n = A + 1/(n+1)$ to obtain $\mathsf{LP}(A)$ as well. Thus $A \in \mathcal{S}$ and $\mathcal{S} = [A, \infty)$. Corollary 3.3 gives reality of every zero at and after A.

Given $\varepsilon > 0$, put $T = A - \varepsilon/2$. Since $T \notin \mathcal{S}$, some $s \geq T$ fails $\mathsf{LP}(s)$. Necessarily $s < A$, since all times at or after A satisfy it. Corollary 3.3 supplies a nonreal zero at this $s \in (A - \varepsilon, A)$.

Choose $s_n \in (A - 1/(n+1), A)$ with nonreal zero z_n. All these times exceed one because $A > 8$. Corollary 9.2 puts all z_n in one compact disk. After passing to a subsequence, $z_n \to w$ and $s_n \to A$. Joint continuity gives $H_A(w) = 0$, and threshold reality gives $w \in \mathbb{R}$.

Suppose $c = H_A'(w) \neq 0$. Joint derivative continuity provides a time neighborhood of A and a disk about the real w in which $|H_s'(q) - c| \leq |c|/2$. Integration along any segment in that convex disk yields

$$|H_s(q) - H_s(p)| \geq \tfrac{1}{2}|c|\,|q - p|.$$

For large n, both z_n and $\overline{z}_n$ lie there and are distinct zeros of the same H_{s_n}. This contradicts the lower bound. Hence $H_A'(w) = 0$. Since $H_A(0) > 0$, $w \neq 0$. If $w < 0$, replace every z_n by $-z_n$ and w by $-w$; evenness preserves all the zero identities, and its derivative is odd. Thus the limit is a positive real multiple zero. Its order is at least two and finite, because H_A is a nonzero entire function. Global eventual simplicity was already proved in Proposition 9.1. □

11 Complete divisor polynomials and Gaussian integration

The polynomial construction used above has a broader classical domain. Keeping its exact zeros also clarifies what approximation alone can and cannot justify. Throughout this section let f be an even entire function, real on the real axis, with $f(0) \neq 0$ and

$$|f(z)| \leq Ce^{|z|^{3/2}} \qquad (z \in \mathbb{C}). \tag{29}$$

Real on the real axis implies $f(\overline{z}) = \overline{f(z)}$ by the identity theorem. No reality or simplicity of its zeros is assumed. For each $R > 0$, choose one representative a of every pair $\{a, -a\}$ of zeros in $|a| < R$, and retain its multiplicity m_a. Define

$$N_R(z) = \prod_{\{a,-a\},\ |a|<R} (1 - z^2/a^2)^{m_a}.$$

This is the complete disk divisor product, not a product that discards imaginary-axis or nonreal zeros. The choice within each pair has no effect.

Proposition 11.1. *There are zero-free boundary radii $R_n \in (2n + 2, 2n + 3)$ such that the real even polynomials $P_n = f(0)N_{R_n}$ converge locally uniformly to f. They satisfy $P_n(0) = f(0)$, and their zeros, with multiplicity, are exactly the zeros of f in $|z| < R_n$. In particular, real coefficients do not mean that the polynomials split over $\mathbb{R}$.*

Proof. There are only countably many excluded zero moduli. In the proof of Lemma 3.1, the finite disk factorization, even normalized logarithm and double-zero Schwarz estimate use evenness and (29), with no real-zero assumption. The same Jensen argument gives $n_f(R) = O(1 + R^{3/2})$ for the total number of zeros, including all complex zeros and multiplicities. The paired denominator error on $|z| \le r$ is therefore at most

$$\exp\left(\frac{n_f(R)r^2}{2R^2}\right) - 1 \longrightarrow 0.$$

Together with $h_R/h_R(0) \to 1$, the exact reconstruction in that proof gives $f(0)N_R \to f$ along the chosen radii. Every factor has roots exactly $a, -a$, with the stated multiplicity, so there are no omitted or added roots. Conjugation permutes the complete zero multiset, hence fixes every coefficient of N_R; those coefficients are real. Its value at zero is one. These observations prove every assertion. □

The result is a concrete classical factorization consequence, consistent with Hadamard theory; McMullen's primary course notes state that theory as Theorem 3.15 and illustrate cancellation of opposite exponential factors in Theorem 3.18 [4]. Our preceding disk argument provides its needed proof here. The corresponding frozen modules are `EvenEntireCanonicalApproximation` and `EvenEntirePolynomialApproximation`.

A common quadratic budget, with an explicit zero strip

Suppose additionally that every zero satisfies $|\operatorname{Im} a| \le b$ for a fixed $b \ge 0$. This hypothesis is retained in the named formal budget theorem. Write

$$S_n = \sum_{\{a,-a\},\ |a|<R_n} \frac{m_a}{a^2}.$$

Differentiating the finite product at zero gives $P_n''(0) = -2f(0)S_n$. Cauchy's integral formula on any fixed circle passes the compact-uniform limit through two derivatives. Thus

$$S_n \longrightarrow -\frac{f''(0)}{2f(0)}. \tag{30}$$

If $|a| \geq 2b$, then $2|\operatorname{Im} a| \leq |a|$, and direct computation yields

$$|a^{-2}| \leq 2\operatorname{Re}(a^{-2}), \qquad \operatorname{Re}(a^{-2}) = \frac{(\operatorname{Re} a)^2 - (\operatorname{Im} a)^2}{|a|^4}.$$

Indeed the numerator is $|a|^2 - 2(\operatorname{Im} a)^2 \geq |a|^2/2$. For $j \geq i$ and $R_i \geq 2b$, this gives the absolute annular estimate

$$\sum_{R_i \leq |a| < R_j} \frac{m_a}{|a|^2} \leq 2\operatorname{Re}(S_j - S_i) \longrightarrow 0 \quad \text{uniformly as } i, j \to \infty. \tag{31}$$

Here and below sums use one member of each opposite pair. Our chosen radii are strictly increasing. Formula (30) makes the right side small; the cone inequality also proves its nonnegativity.

Theorem 11.2. *For every $\epsilon > 0$ there is $C_\epsilon < \infty$ such that*

$$|P_n(z)| \leq C_\epsilon e^{\epsilon|z|^2} \qquad \textit{for every } n \geq 0 \textit{ and } z \in \mathbb{C}.$$

The constant is common to all complete divisor approximants, although it may depend on f and ϵ.

Proof. Choose i so large that every later annulus in (31) has sum at most $\epsilon/2$. Since $|1-w| \leq e^{|w|}$, its product has modulus at most $\exp(\epsilon|z|^2/2)$. The fixed inner polynomial has a bound $Ce^{\epsilon|z|^2/2}$: each of its finitely many factors satisfies

$$|1 - cz^2| \leq (1 + |c|/\delta)e^{\delta|z|^2}, \qquad \delta > 0.$$

To see this, use $1+|c||z|^2 \leq (1+|c|/\delta)(1+\delta|z|^2)$ and $1+x \leq e^x$. Allocate $\delta = (\epsilon/2)/(1+M)$, where M is the total inner pair multiplicity. Multiplying its factors proves the inner bound. Factor P_n into inner and annular products for $n \geq i$, including its fixed factor $f(0)$. The two budgets add to ϵ. Each of the finitely many earlier polynomials has its own such bound; taking their maximum completes the proof for all n. □

These are the actual mechanisms in the two frozen modules:
`EvenEntireInverseSquareTails`;
`EvenEntireGaussianBudget`.
The strip is a hypothesis about this entire function's zeros; it is not an assumption that they are real.

What now passes through an actual Gaussian integral

Let Y have any fixed real Gaussian probability law, possibly degenerate, and fix $z, c \in \mathbb{C}$. Then

$$\mathbb{E}|P_n(z + cY) - f(z + cY)| \longrightarrow 0, \qquad \mathbb{E}P_n(z + cY) \longrightarrow \mathbb{E}f(z + cY). \tag{32}$$

All these random variables are genuinely integrable. For a direct proof, choose $a > 0$ with $\mathbb{E}e^{aY^2} < \infty$, available for every real Gaussian law (and immediate for a constant law). Choose $\epsilon = a/(4(1 + |c|^2))$. The inequality $|z + cy|^2 \leq 2|z|^2 + 2|c|^2y^2$ gives the common majorant

$$|P_n(z + cy)| \leq C_\epsilon e^{2\epsilon|z|^2} e^{ay^2}.$$

The same bound holds for f by pointwise convergence. Twice this integrable majorant controls the difference, so dominated convergence proves both limits. This is the fixed-affine Gaussian L^1 conclusion actually mapped in `gaussian_L1_limit`; it does not invoke a formal interchange of an unbounded series of differential operators.

For clarity, a further classical *human deduction* is local uniformity in the affine parameters: if $|z| \le A$, $|c| \le B$, choose $\epsilon = a/(4(1+B^2))$ once. For each fixed y, local uniformity of $P_n \to f$ gives uniform convergence over that compact parameter set, because its affine images lie in $|w| \le A + B|y|$. The supremum of the continuous error over the compact parameter set is measurable in y (for example use a countable dense subset). The same common majorant and dominated convergence show that the supremum of the expected absolute errors tends to zero. This strengthens only the parameter quantifiers of (32), not the class of clocks.

Compact-uniform convergence alone would be insufficient. A precise human counterexample for standard Y is

$$Q_n(z) = 1 + \frac{z^{2n}}{(2n-1)!!}, \qquad n \ge 1.$$

Since $(2n-1)!! \ge n!$, on $|z| \le A$ its added term is at most $A^{2n}/n! \to 0$ (the ratio tends to zero). Hence $Q_n \to 1$ locally uniformly, with $Q_n(0) = 1$, evenness and real coefficients. Nevertheless $\mathbb{E}Q_n(Y) = 2$ for every n, whereas $\mathbb{E}1 = 1$; the expected absolute error is exactly one. These are not asserted to be complete divisor products of a common function or to have a fixed zero strip. Their role is to disprove an interchange based solely on compact convergence.

Finally, a Gaussian affine average is a quadratic smoothing operation. Even when c is complex and the polynomial average can be written as a finite differential sum, the proof above is an integral domination proof. It neither identifies a growing quartic deformation with Gaussian smoothing nor propagates real zeros under that quartic deformation. The attained threshold and every limitation of the preceding sections therefore retain exactly their original scope.

12 Formal correspondence and limits of the conclusion

The mathematical baseline is the frozen verified commit

`bab688bb6140cd6316c80680095a914ea0d346db`.

The accompanying source dossier records exact declaration types, lines, hashes and links, the finite certificate table, and its independent rational aggregation check. The main formal namespaces, with prefix `OperatorDerivativeRewrite.Foundation`, are:

`RiemannQuarticUniformFarScale`
`RiemannQuarticGlobalSimpleZeros`
`RiemannQuarticActualPolynomialLimit`
`RiemannQuarticThresholdEndpoint`
`RiemannQuarticThresholdZeros`
`RiemannQuarticThresholdCompactness`
`RiemannQuarticThresholdCollision`.

The actual certificate is `actual_real_derivative_negative` in `RiemannQuarticLaguerreWitness`.

The human proof groups the actual logarithmic-derivative estimates into one admissible-saddle calculation and uses unspecified uniform constants. The formal source retains explicit constants, model corrections and separate transition and far budgets. The regrouping is an exposition of the

same final result, not an assertion that every intermediate lemma above has a new named Lean declaration. The exact final quantifiers are those of the mapped declarations. The arithmetic finite inclusions retain their explicit computer-assisted proofs, while the analytic regrouping and polynomial reconstruction are supplied as human proofs here.

The threshold is an attained eventual-start threshold. Neither monotone propagation from each individual reality time nor failure at every earlier time has been proved here. The collision argument proves finite multiplicity at least two, not exactly two or a specified zero trajectory. The upper threshold is existential, without a numerical upper value or an optimal quantitative estimate. No comparison with the classical quadratic threshold or implication for the Riemann hypothesis is supplied.

De Bruijn [1] and Newman [2] are primary antecedents for real-zero Fourier transforms and quadratic propagation. Their exact deformation, sign and quantifiers must be retained in any priority comparison. Preliminary comparison does not establish novelty for the combined quartic package; novelty remains unresolved.

References

[1] N. G. de Bruijn, The roots of trigonometric integrals, *Duke Math. J.* **17** (1950), 197–226. doi:10.1215/S0012-7094-50-01720-0.

[2] C. M. Newman, Fourier transforms with only real zeros, *Proc. Amer. Math. Soc.* **61** (1976), 245–251. `https://sites.math.northwestern.edu/~auffing/papers/Newman.pdf`.

[3] NIST Digital Library of Mathematical Functions, §20.7(viii), equation 20.7.32, modular theta identity. `https://dlmf.nist.gov/20.7.E32`.

[4] C. McMullen, *Advanced Complex Analysis*, Harvard Math 213a course notes, November 30, 2023, Theorems 3.15 and 3.18, printed pages 80–82. `https://people.math.harvard.edu/~ctm/home/text/class/harvard/213a/23/html/home/course/course.pdf`.